

Clinical Risk Assessment Report: Patient 91

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Borderline / Low Confidence
- Absolute Predicted Risk: 21.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.98x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 4.7%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Clinical Sensitivity Analysis (Borderline Patient):

PREDICTION INSTABILITY WARNING: This patient's risk profile sits precisely on the borderline. Consequently, the model's prediction is highly sensitive. Minor biological variations or measurement errors in the following features would flip the final clinical prediction:

- Small Zone Emphasis (PET): A decrease would alter the prediction outcome.
- 75th Percentile Tumor Density: A decrease would alter the prediction outcome.
- Large Zone Emphasis (CT): A decrease would alter the prediction outcome.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Patient Age (Age) **Value: 52.0 [Bottom 15%] (+3.2%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.82 [Top 21%] (+3.0%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 38.47 [Top <1%] (+2.4%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 3.1 [Top <1%] (+2.3%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.45 [Avg (71th %ile)] (+1.6%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.13 [Avg (41th %ile)] (-4.7%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

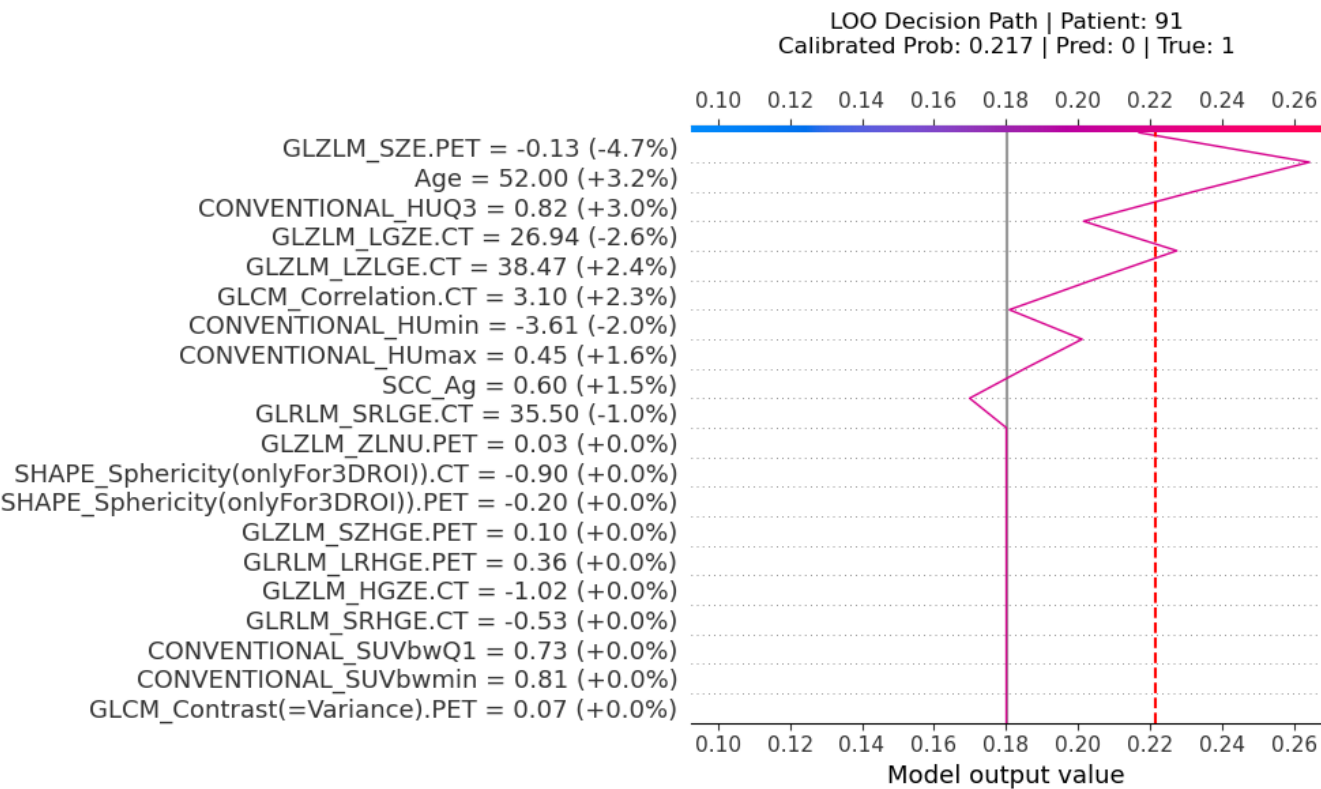
Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 26.94 [Top <1%] (-2.6%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Minimum Tumor Density (CONVENTIONAL_HUmin)

Value: -3.61 [Bottom <1%] (-2.0%)

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

